



ACADEMIC REGULATIONS & SYLLABUS

Faculty of Computer Science & Applications
Smt. Chandaben Mohanbhai Patel Institute of
Computer Applications
M.Sc.(IT) Programme





CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Computer Science & Applications

ACADEMIC REGULATIONS

M.Sc. (IT) Programme
(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2018-2019

CHARUSAT

FACULTY OF COMPUTER SCIENCE AND APPLICATIONS

ACADEMIC RULES

Master of Science in Information Technology(M.Sc. (IT).) Programme

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. Duration of Programme

- | | |
|-----|---|
| (i) | Postgraduate programme : Master of Science in Information Technology |
| | Minimum 4 semesters (2 academic years) |
| | Maximum 6 semesters (3 academic years) |

3. Eligibility & Mode of admissions

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Government of Gujarat from time to time.

4. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

5. Attendance

- 5.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected

Faculty of Computer Science and Applications

absence from classes and other activities will be required to be condoned by the Dean/Principal.

5.2 Student attendance in a course should be 80%.

6 *Course Evaluation*

6.1 The performance of every student in each course will be evaluated as follows:

- 6.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and
- 6.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

6.2 *University Examination*

- 6.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.
- 6.2.2 In order to earn the credit in a course a student has to obtain grade other than FF.

6.3 Performance at Internal & University Examination will be done on the relative grading system.

7 *Grading*

7.1 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are defined as follows:

(i) $SGPA = \sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester

(ii) $CGPA = \sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

- (iii) Students having CGPA of at least 3.0 will be allowed to go to 3rd semester (i.e. Second year)
- (iv) A) Students having backlogs of 2 or less together in 1st and 2nd semester (i.e. First year) and /or backlogs of 4 or less together in 3rd and 4th semesters (i.e. Second year)
And
B) Students having CGPA of at least 4.0 will be allowed to go to 5th semester (i.e. Third year)
- (v) In addition to above, the student has to complete the required formalities as per the regulatory bodies.

8 Awards of Degree

8.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

- 8.1.1 He should have earned at least minimum required credits as prescribed in course structure; and
- 8.1.2 He should have cleared all evaluation components in every course; and

8.1 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for the award of degree. Only latest grade will be considered.

9 Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	CGPA ≥ 7.5 & ≤ 10
First class:	CGPA ≥ 6.0 & < 7.5
Second Class:	CGPA ≥ 5.0 & < 6.0
Pass Class:	CGPA < 5.0

10 Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)

FACULTY OF COMPUTER SCIENCE & APPLICATIONS

CHOICE BASED CREDIT SYSTEM

FOR

MASTER OF SCIENCE (INFORMATION TECHNOLOGY)

CHOICE BASED CREDIT SYSTEM

The choice based credit system provides flexibility in designing curriculum and assigning credits based on the course content and hour of teaching. The choice based credit system provides an opportunity for the students to choose courses from the prescribed courses comprising core, elective and open elective courses. The CBCS provides a cafeteria type approach in which the students can take courses of their choice and adopt an interdisciplinary approach to learning. The courses shall be evaluated on the grading system, which is considered to be better than the conventional marks system.

CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 3 types of courses:

Foundation Courses, Core Courses and Elective Courses.

Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core Courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme.

B. Programme Core Courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialized or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes.

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialization.

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

Faculty of Computer Science and Applications

Vision

To become a leading institution in the field of computer applications and contribute in national efforts of computerizing public systems

Mission

To produce competent computer professionals with the ability to face future challenges.

Program Educational Objectives (PEOs)

Graduates of Master of Computer Science in Information Technology (M.Sc.(IT)) program at the Charotar University of Science and Technology (CHARUSAT) should be able to utilize the knowledge gained from Their academic program to:

- PEO1 Be successful professionals in industry, government, research, entrepreneurial pursuit and consulting firms.
- PEO2 Integrate fundamentals and advanced concepts of computer science and applications to develop innovative solutions to societal problems with effective usage of software development methodologies.
- PEO3 Develop competency to be placed as per the current market trends and requirements.
- PEO4 Prepare individual in a manner that he/she can be offered job of versatility.
- PEO5 Be sensitive to the consequences of their work, both ethically and professionally, for productive professional careers.

PROGRAMME OUTCOMES (POs):

The curriculum of Master of Computer Science in Information Technology (M.Sc.(IT)) program at the Charotar University of Science and Technology (CHARUSAT) is designed to assure students, at the time of graduation are able to:

- PO1. Ability to think out of box for the effective and robust software application development.
- PO2. To deal with constraints and other barricades in the area of software application development.
- PO3. Effective usage of various ICT tools for the advanced software development.
- PO4. To be competitive in the fields of research and innovations.
- PO5. To become effective communicator to satisfy need of industry in non-technical IT profile.
- PO6. Function effectively within multidisciplinary teams.
- PO7. Provide solutions for various societal problems with automation of systems.
- PO8. Work with a professional attitude and moral responsibility.

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Teaching and Examination Scheme (M.Sc(IT) Programme) Effective from Year 2018-19

SEMESTER-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theo ry	Pract	Total		Internal		Exter nal	Internal		Exter nal	
						Case Study	Tests		Term work	Tests		
MS312-315	Elective-III	3	-	3	3	10	20	70	-	-	-	100
MS316	Web Designing	-	3	3	3	-	-	-	15	15	70	100
MS317	Web Development Using Advanced Java	4	3	7	7	10	20	70	15	15	70	200
MS318	Web XML Applications	4	3	7	7	10	20	70	15	15	70	200
MS319	Web Development Using Open Source	4	3	7	7	10	20	70	15	15	70	200
CL142 A	Environmental Sciences	2	-	2	2	30		70	-		-	100
Total		17	12	29	29	500			400			900

Elective-III

MS312-Statistical Methods and Optimization Techniques

MS313-Advanced Operating Systems

MS314-Cyber Security Fundamentals

MS315-Basics of Cloud Computing

Teaching and Examination Scheme (M.Sc(IT) Programme) Effective from Year 2018-19

SEMESTER-IV

Course Code	Course Title	Teaching Scheme				Internal	End Semester Examination		Total
		Contact Hours			Credit	Continuous Evaluation	Report	Presentation & Viva	
		Inst.	Industry	Total					
MS401	Dissertation/Project Work	2	28	30	30	200	200	400	800

TEACHING SCHEME & DETAILED SYLLABUS

FOR

**M.Sc. (IT) PROGRAMME
(3rd SEMESTER)**

**EFFECTIVE FROM
ACADEMIC YEAR 2018-19**

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SEMESTER-III

Course Code	Course Title	Teaching Scheme				Examination Scheme						
		Contact Hours			Credit	Theory			Practical			Total
		Theo ry	Pract	Total		Internal		Exter nal	Internal		Exter nal	
						Case Study	Tests		Term work	Tests		
MS312-315	Elective-III	3	-	3	3	10	20	70	-	-	-	100
MS316	Web Designing	-	3	3	3	-	-	-	15	15	70	100
MS317	Web Development Using Advanced Java	4	3	7	7	10	20	70	15	15	70	200
MS318	Web XML Applications	4	3	7	7	10	20	70	15	15	70	200
MS319	Web Development Using Open Source	4	3	7	7	10	20	70	15	15	70	200
CL142 A	Environmental Sciences	2	-	2	2	30		70	-		-	100
Total		17	12	29	29	500			400			900

Elective-III

MS312-Statistical Methods and Optimization Techniques

MS313-Advanced Operating Systems

MS314-Cyber Security Fundamentals

MS315-Basics of Cloud Computing

**MS312: Statistical Methods and Optimization Techniques
(100 Marks)**

Credit Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: During Theory lectures illustrations of certain real world problems requiring Statistical inference will be explained. Statistical methods will be introduced as tools for solving such problems also are emphasized.

Learning Outcomes: Upon successful completion of the course, the students will learn advance concepts of statistics. Also, they will be able to apply optimization techniques to construct suitable optimization model to solve real life strategic problems – issues.

Outline of the Course:

Unit No.	Title of the Unit	Minimum No. of Hours
		Theory
1	Probability Distribution	06
2	Sampling Distribution & Interval Estimation	07
3	Statistical Inference - Testing of Hypothesis	07
4	Basics of Operations Research and Linear Programming	05
5	Assignment Problem & Transportation Problem	06
6	Activity Network Analysis	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Probability Distribution Hours: 06

Probability Distribution: Discrete Probability Distributions; Expected Value & Variance; Binomial Probability Distribution, Poisson Probability Distribution, Normal Probability Distribution, Normal Approximation of Binomial Probabilities, Exponential Probability Distribution

Unit – II: Sampling Distribution & Interval Estimation Hours: 07

Simple Random Sampling, Point Estimation, Properties of Point Estimation, Other Sampling Methods, Population Mean: Known, Unknown, Determining the Sample Size; Population Proportion, The method of moments and the method of maximum likelihood estimation, Interval Estimation Methods viz. Confidence Intervals for Mean of the Normal Distribution with known and unknown variance, Confidence Interval for variance of the Normal Distribution, Confidence Intervals for Difference of Two Means, Sample Size.

Unit – III: Statistical Inference - Testing of Hypothesis Hours: 07

Introduction to Hypothesis Testing: Null Hypothesis, Alternate Hypothesis, Type-I & Type-II Errors; Testing of Hypotheses about a Population Mean using the z-statistic (Sigma Known); Testing of Hypotheses about a Population Mean using the t-statistic (Sigma Unknown); The Gamma and Chi-square, t, z and F tests ,Testing of Hypotheses about a Proportion; Testing of Hypotheses about a Variance; Hypotheses Testing and Confidence Intervals about the Difference in Two Means using the z-statistic (Sigma Known); Hypotheses Testing and Confidence Intervals about the Difference in Two Means: Independent Samples (Sigma Unknown); Statistical Inferences for Two Related Populations; Statistical Inferences About Two Population Proportions, $p_1 - p_2$; Testing of Hypotheses About Two Population Variances

Unit – IV: Basics of Operations Research and Linear Programming Hours: 05

Overview of Operations Research: Features and Advantages of Operation Research.
Linear Programming: Structure and mathematical model of linear programming Problem, Linear Programming Model Formulation: Graphical Method, Simplex method

Unit – V: Assignment Problem & Transportation Problem Hours: 06

Transportation Problem: Mathematical model of Transportation Model, Method for finding initial solution: NWCM method, LCM method, Vogel's Approximate method and MODI method.
Assignment Problem: Mathematical model of assignment problem, Solution method of assignment problem- Hungarian method.

Unit – VI: Activity Network Analysis Hours: 05

Introduction, Basic Difference Between PERT and CPM, Phases of Project Management, PERT/CPM Network Components & President Relationship, Critical Path Analysis, Project Scheduling with Uncertain Activity Times.

Core Books:

1. J K Sharma: Operations Research – Theory and Applications, 4th Edition, MacMillan Business Book, 2011.

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2. S P Gupta: Statistical Methods, 30th Edition, S Chand Publications, Delhi, 2014.

Reference Books:

1. S. P. Gordon & F. S. Gordon: Contemporary Statistics- A Computer Approach, McGraw Hill International Edition, 1994.
2. Hogg & Tannis: Probability and Statistical Inference, 6th Edition, Prentice-Hall, NJ, USA, 2000.
3. Larsen & Marx: An Introduction to Mathematical Statistics and Its Applications, 3rd Edition, Prentice Hall, NJ, USA, 2001.
4. Shah, Gor, Soni, "Operations Research", PHI, 2008.

Web References:

1. <http://nptel.ac.in/courses/112106134/1> [Notes]
2. <http://nptel.ac.in/downloads/111104032/> [Notes]

Course Outcomes: Upon successful completion of the course, the students will be able to:

- C01 : Understand the concepts of probability distribution.
C02 : Learn Interval Estimation Methods in context to real applications.
C03 : Understand Hypothesis Testing for any research problem.
C04 : Understand Operation Research ,Linear Programming and assignments and transportation problems.
C05: Learn Network Analysis with PERT and CPM.

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Probability Distribution	✓				
2	Sampling Distribution & Interval Estimation		✓			
3	Statistical Inference - Testing of Hypothesis			✓		
4	Basics of Operations Research and Linear Programming				✓	
5	Assignment Problem & Transportation Problem				✓	
6	Activity Network Analysis					✓

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Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	-	-	2	-	-	1	-
CO2	2	2	-	2	-	-	1	-
CO3	-	3	-	2	-	2	-	-
CO4	3	-	-	2	2	-	1	2
CO5	2	3	-	2	3	3	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS313: Advanced Operating Systems (100 Marks)

Credit Hours: 03

Pre-requisite: MS112: Operating Systems concepts

Methodology & Pedagogy: During theory lectures the emphasis will be given on the advanced concepts of Operating Systems. Students will be introduced jargons of distributed and mobile based operating systems. During theory lectures concepts of shared memory, remote procedure call, synchronization, process management, resource management and distributed file systems will be discussed. Students will also give overview of mobile based operating systems.

Learning Outcomes: Upon successful completion of the course, the students will learn advanced concepts of operating systems in terms of distributed environment. Students will learn functionalities of operating systems in distributed environment.

Outline of the Course:

Unit No.	Title of the Unit	Minimum No. of Hours
		Theory
1	Overview of Distributed Systems	04
2	Distributed Shared Memory and Remote Procedure Calls	08
3	Synchronization in Distributed Operating Systems	06
4	Process and Resource Management in Distributed Operating Systems	08
5	Distributed File Systems	05
6	Mobile and Real time Operating Systems	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Overview of Distributed Systems Hours: 04

What is Distributed operating system, issues in designing distributed operating system – transparency, Reliability, Flexibility, performance, scalability, heterogeneity, security, Emulation of existing operating system, Introduction to distributed computing environment.

Unit – II: Distributed Shared Memory and Remote Procedure Calls Hours: 08

Introduction to RPC, RPC Model, Implementing RPC Mechanism, Stub Generation, RPC Message, Introduction to Distributed Shared Memory, General Architecture of DSM Systems, Design and implementation issues of DSM, Granularity, Structure of Shared memory space.

Unit – III: Synchronization in Distributed Operating Systems Hours: 06

Clock Synchronization – Implementation, drifting of clocks, Clock synchronization issues, Clock Synchronization algorithms, Mutual Exclusion – Centralized approach, Distributed Approach, Token Passing Approach.

Unit – IV: Process and Resource Management in Distributed Operating Systems Hours: 08

Introduction to resource management in distributed operating system, Desirable features of good global scheduling algorithm, Task Assignment approach, Load balancing Approach, Load Sharing Approach, Process migration – desirable feature of Process migration, process migration mechanisms, Process migration in heterogeneous systems, Advantages of process migration.

Unit – V: Distributed File Systems Hours: 05

What is Distributed File system – Remote information sharing, User mobility, Availability, Diskless workstation. Types of services in distributed file system – Storage service, True file Service, Name Service. Desirable features of Distributed file system, File Models, File accessing models.

Unit – VI: Mobile and Real Time Operating Systems Hours: 05

Introduction to Mobile Phone Systems, Scheduling in Symbian OS, Concurrency in Symbian OS, Memory in Symbian OS, I/O in Symbian OS, File system in Mobile Phones, Symbian OS message model structure, Overview of Real time OS.

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Core Books:

1. Pradip K. Sinha: Distributed Operating Systems Concepts and Design, Eastern Economy Edition, PHI, 2007.
2. Michael J. Jipping : Smartphone Operating System Concepts with Symbian OS, Wiley, 2007.

Reference Books:

1. Andrew S. Tanenbaum: Distributed Operating Systems, 5th Edition, Pearson, 2008.
2. Doreen L. Galli: Distributed Operating Systems Concepts and Practice, Prentice Hall, 2000.
3. Andrew S. Tanenbaum, Herbert Bos: Modern Operating Systems, 4th Edition, Pearson Education, 2014.

Web References:

1. www.cs.columbia.edu/smb/classes/s06-4118/l26.pdf [Overview of Distributed OS]
2. www.cs.uah.edu/~weisskop/Notes690/A5_DistSysCh1.ppt [PPTs of Distributed OS]

Course Outcomes: Upon successful completion of the course, the students will :

- C01 : Be familiar with Distributed Operating Systems and their designing issues
- C02 : Be familiar with the functionalities of Distributed Operating Systems
- C03 : Be Familiar with Mobile and Real Time Operating systems with their functionalities
- C04 : Be able to design the functionality of advanced operating systems
- C05 : Be able to perform independent research in distributed, mobile and real time Operating Systems.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Overview of Distributed Systems	✓			✓	✓
2	Distributed Shared Memory and Remote Procedure Calls	✓	✓		✓	✓
3	Synchronization in Distributed Operating Systems	✓	✓		✓	✓
4	Process and Resource Management in Distributed Operating Systems	✓	✓		✓	✓

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5	Distributed File Systems	✓	✓		✓	✓
6	Mobile and Real time Operating Systems			✓	✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	2	2	3	1	-	2	1
CO2	3	2	2	3	1	-	2	1
CO3	2	2	2	3	1	-	2	1
CO4	2	2	2	3	1	-	2	1
CO5	3	2	2	3	1	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS314: Cyber Security Fundamentals (100 Marks)

Credit Hours: 03

Pre-requisite: Basic knowledge of fundamental Networking.

Methodology & Pedagogy: During theory lectures the emphasis will be given on the basics of cyber-crime, tools and techniques used in cyber-crime, devices used to perform cyber-crime, detection and prevention of cyber-crimes. The laws prevailing for cyber-crimes are also discussed to know the implications of cyber-crime. Cybercrime real life, Examples and Mini-Cases and online scams will be discussed to enhance understanding.

Learning Outcomes: Upon successful completion of the course, the students will learn what is cyber-crime, classification of cyber-crime, methods used to perform crime, apply cyber security, and know the detailing of Information Technology Acts against offences.

Outline of the Course:

Unit No.	Title of the Unit	Minimum No. of Hours
		Theory
1	Introduction to Cyber Crime	05
2	Tools and Techniques Used in Cyber Crime	07
3	Wireless Device and Cyber Crime	06
4	Cyber Security	07
5	Detection and Prevention of Cyber Crime	06
6	Provisions in Indian Laws in dealing with Cyber Crimes	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Introduction to Cyber Crime Hours: 05

Introduction to cyber crimes , Distinction between cyber crime and conventional crimes, Reasons for commission of cyber crime, Common Cyber Threats
Classification Of Cyber Crimes: Cyber crime against Individual, Cyber crime Against Property, Cyber crime Against Organization, Cyber crime Against Society

Unit – II: Tools and Techniques Used in Cyber Crime Hours: 07

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Cyber offenses: Cyber criminals, organized cyber crimes, Types of attacks, Cybercafé and Cybercrimes, Botnet, Botnet Proxy, Introduction to Cloud Computing, Types of Services, Cybercrime and Cloud Computing.

Methods Used for Cybercrime: Proxy Servers and Anonymizers, Password Cracking, Phishing, Key loggers, Antikey logger, Spywares Trojan

Attacks on Wireless Networks: Traditional Techniques of Attacks on Wireless Networks, Theft of Internet Hours and Wi-Fi-based Frauds and Misuses.

Unit – III: Wireless Device and Cyber Crime Hours: 06

Introduction: Proliferation of Mobile and Wireless Devices, Mobile Phone Theft, Mobile Viruses, Mishing, Vishing, Smishing, Hacking Bluetooth

Credit Card Frauds in Wireless Device: Types and Techniques of Credit Card Frauds, Security Challenges Posed by Mobile Devices, Protecting Data on Lost Devices.

Unit – IV: Cyber Security Hours: 07

Cyber Security (IT security), Security principles, Security triad: Confidential, Integrity, Availability, Security Policy, Security Service Life Cycle

Aspects of Organizational Security- Information Security, Information Security's Overview and Services, Physical security, E-commerce Security, Legal security, Email security, Goals of Security.

Unit – V: Detection and Prevention of Cyber Crime Hours: 06

Cyber security techniques: Steganography, Digital Evidence Collection, Browser Forensic, Wireless Network Scanner, Disk Cloning, Internet History Security, Internet Evidence Finder

Identity Theft (ID Theft): Personally Identifiable Information (PII), Types of Identity Theft, Techniques of ID Theft, Identity Theft-Countermeasures, How to Protect your Online Identity.

Unit – VI: Provisions in Indian Laws in dealing with Cyber Crimes Hours: 05

Information Technology Act, 2000, Penalties Under IT Act, Offences Under IT Act, Offences Related with Digital Signature and Electronic Signature Under IT Act, Statutory Provisions, Establishment of Authorities under IT Act and their functions, powers, etc., Controller, Certifying Authorities, Cyber Regulation Appellate.

Core Books:

1. Nina Godbole, Sunit Belapur: Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives, Wiley India Publications, April, 2011.
2. Robert Jones: Internet Forensics - Using Digital Evidence to Solve Computer Crime, O'Reilly Media, October, 2005.
3. Farooq Ahmad: Cyber Law in India, Pioneer Books, 2001.

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Reference Books:

1. Chad Steel: Windows Forensics - The field guide for conducting corporate computer investigations, Wiley India Publications, December, 2006.
2. Eoghan Casey: Digital Evidence and Computer Crime, 3rd Edition, Academic Press, 2011.

Web References:

1. <http://www.lawyersclubindia.com/articles/Classification-Of-CyberCrimes1484.asp#.VWBGbdKqqko> [Classification Of Cyber Crimes]
2. <http://www.cyberlawclinic.org/cybercrime.htm> [Details regarding Cyber-crime, laws, case studies ,etc.]

Course Outcomes: Upon successful completion of the course, the students will :

- C01 : Be familiar with fundamentals of Cyber Security.
- C02 : Aware about tools and techniques used in Cyber Crime.
- C03 : Understand cybercrime and security in context to wireless device.
- C04 : Understand techniques related to detection and prevention of Cyber Crime.
- C05 : Learn about Laws related to Cyber Crime.

Course Outcomes Mapping :

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to Cyber Crime	✓				
2	Tools and Techniques Used in Cyber Crime		✓			
3	Wireless Device and Cyber Crime			✓		
4	Cyber Security			✓		
5	Detection and Prevention of Cyber Crime				✓	
6	Provisions in Indian Laws in dealing with Cyber Crimes					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	1	2	2	3	-	-	2	3
CO2	2	2	2	3	-	-	2	3
CO3	2	2	2	3	-	-	2	3
CO4	2	2	2	3	-	-	2	3
CO5	1	2	2	3	-	-	2	3

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS315: Basics of Cloud Computing (100 Marks)

Credit Hours: 03

Pre-requisite: MS112: Operating Systems concepts

Methodology & Pedagogy: During theory lectures the emphasis will be given on the fundamentals of cloud computing. Students will be introduced basic types, architecture, service providers, mechanism, security issues and some hidden aspects of cloud computing. Students will give practical exposure in form of case study and by showing cloud infrastructure of university.

Learning Outcomes: Upon successful completion of the course, the students will learn fundamental concepts of cloud computing and business use of it in real time. Student will learn entire architecture of cloud and its business use with security aspects.

Outline of the Course:

Unit No.	Title of the Unit	Minimum No. of Hours
		Theory
1	Evolution of Cloud Computing	04
2	Understanding Cloud Computing and basic types	07
3	Fundamentals Cloud Architecture and Service Providers	07
4	Cloud Computing Mechanisms	07
5	Cloud Computing Security and Business Use	06
6	Hidden Aspects of Cloud Computing	05

Total Hours (Theory): 36

Total: 36

Detailed Syllabus:

Unit – I: Evolution of Cloud Computing Hours: 04

Introduction of Cloud Computing, Growth of technology, Paradigm Shift in Computing, distributed nature of service Provisioning, Support entrepreneurship using Cloud Computing.

Unit – II: Understanding Cloud Computing and basic types Hours: 07

Advantages and drawbacks of Cloud Computing, Essential component for Cloud contract, Major outage of Cloud Computing and Enhancers for Cloud Computing.

Introduction to SaaS, PaaS, IaaS. Introduction to Public Cloud, Private Cloud, Hybrid Cloud and Community Cloud, Storage Services for Cloud Computing

Unit – III: Fundamentals Cloud Architecture and Service Providers Hours: 07

Workload Distribution Architecture, Resource Pooling Architecture, Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture.

Introduction to major Cloud Service Provider: Amazon Web Services, Google Apps.

Microsoft Windows Azure and Office 365, Hp Cloud, RackSpace, CSC Corp, Verizon Terrimark, DropBox.

Unit – IV: Cloud Computing Mechanisms Hours: 07

Introduction to Cloud Infrastructure Mechanisms: Logical Network Perimeter, Virtual Server, Cloud Storage Device, Cloud Storage Levels, Network Storage Interfaces, Object Storage Interfaces, Database Storage Interfaces, Relational Data Storage, on-Relational Data Storage, Cloud Usage Monitor, Monitoring Agent, Resource Agent, Polling Agent, Resource Replication.

Introduction to Cloud Management Mechanisms: Remote Administration System, Resource Management System, SLA Management System, Billing Management System.

Unit – V: Cloud Computing Security and Business Use Hours: 06

Introduction to Encryption, Symmetric Encryption, Asymmetric Encryption, Hashing, Digital Signature, Public Key Infrastructure (PKI), Identity and Access Management (IAM), Single Sign-On (SSO), Cloud-Based Security Groups. Overview of Compliance and Certification, Access Control, Organizational Control.

Benefits of Business using Cloud Computing, Risk of Cloud Computing, Cost factor in Cloud Computing.

Unit – VI: Hidden Aspects of Cloud Computing Hours: 05

Introduction to Hidden Aspects of Cloud Computing, Service level Agreement, Sharing Log Data, Service Uptime Guarantee.

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Core Books:

1. S. Srinivasan: Cloud Computing Basics, Springer, 2014.
2. Thomas Erl, Zaigham Mahmood, and Ricardo Puttini: Cloud Computing Concepts, Technology & Architecture ,PHI, 2013 .

Reference Books:

1. Derrick Rountree, Ileana Castrillo : The Basics of Cloud Computing , Syngress, 2013.
2. Rajkumar Buyya, James Broberg, Andrzej M. Goscinsk: Cloud Computing- Principles and Paradigms, John Wiley & Sons, 2011.

Web References:

1. http://whatiscloud.com/basic_concepts_and_terminology/cloud [For basic terminology of Cloud Computing]
2. http://www.tutorialspoint.com/cloud_computing/ [For cloud computing lecture notes]
3. <http://www.intel.in/content/dam/www/public/us/en/documents/guides/cloud-computing-virtualization-building-private-iaas-guide.pdf> [For cloud computing virtualization]
4. www.cs.purdue.edu/.../Any-Kim-Bhargava-MCCWorkshop.ppt [Security issues PPTs]

Course Outcomes: Upon successful completion of the course:

1. C01 : Students will learn basics of cloud computing.
2. C02 : The students will be familiar with various cloud architectures and services.
3. C03 : They will get the knowledge of various network mechanics.
4. C04 : Students will get various business aspects of cloud computing with security aspects.

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes			
		C01	C02	C03	C04
1	Evolution of Cloud Computing	✓			
2	Understanding Cloud Computing and basic types		✓		
3	Fundamentals of Cloud Architecture and Service Providers		✓		
4	Cloud Computing Mechanisms			✓	
5	Cloud Computing Security and Business Use				✓
6	Hidden Aspects of Cloud Computing				✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	-	-	1	-	1	1	1	-
CO2	3	2	3	1	1	1	1	-
CO3	1	-	2	2	-	-	1	1
CO4	1	1	1	2	1	1	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS316: Web Designing (100 Marks)

Credit Hours: 03

Pre-requisite: None.

Methodology & Pedagogy: The course will cover the usage of graphics and discuss the appropriateness of multimedia on Web sites. Students will learn to design layouts for web pages, paper adverts, brochures, event & exhibition stall designs and to create animated graphics. We'll take a look at design issues specific to web-based presentations, learn web page layout, effective navigation and delve into the design process. We examine some of the benefits and pitfalls of using graphics, color and fonts on web pages as well as working with several small Web projects.

Learning Outcomes: Upon successful completion of the syllabus students shall be able to understand the foundation of web site design, Issues of diversity and inclusiveness, working with Photoshop, Using Flash, create web sites that are functional and well designed and engaging multimedia to accompany sites.

Outline of the Course:

Week No.	Content
1	Getting Started with Photoshop and Working with layers. Introduction Photoshop Interface, Image size and resolution , creating new images , Opening, importing & placing images , Introduction to Tool Box & Tool Option bar , Overview of layers and Blending Modes
2	Selections Tool & Transformation Introduction, Selecting images, Cropping images ,Transforming Objects
3	Retouching Tool Retouching the Image, Healing Brush Tool, Clone Stamp Tool, Pattern Stamp Tool, Eraser Tool
4	Painting Tool and Working with text Painting tools available in Photoshop , Brush Presets, type tool
5	Creating Shapes Painting ,Drawing ,Pen tool ,Work path
6	Masking, Adjustment and other Tools
7	Filters and Animation in PHOTOSHOP
8	Getting Started with Flash Introduction to workspace , Organizing Panels , Working with a Document , Working with Rulers, Grid and Guides , Working with Time-line, Working with tools, Testing and Publishing animated file
9	Creating Animation Creating Frame-by-Frame Animation, Classic tweening, Shape tweening and Motion tweening

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10	Create Motion Guide and Masking in animation
11	Introduction to Action Script 3.0 Array and Loops, Controlling Actions with Events, Organizing Objects with the Display List
12	Drawing of objects and Event handling

Total Hours (Lab): 36

Total: 36

Core Books:

1. Lesa Snider: Photoshop CS5 - The missing manual, O'reilly Media, 1st Edition, May 2010.
2. Lisa DaNae Dayley, Brad Dayley: Photoshop CS5 - Adobe Photoshop CS5 Bible, Wiley India Pvt. Ltd., June 2010.
3. Chris Grover: Flash CS5, Action Script 3.0 - The Missing Manual, 1st Edition, O'Reilly Publication.

Reference Books:

1. Katherine Ulrich: Flash Professional CS5, Peachpit Press.
2. Michael Toot, Sherry Kinkoph: Master Adobe Photoshop, Illustrator, Premiere and after effects Visually, 1st Edition 2002.
3. Douglas K. Van Duyne, James A. Landay, Jason I. Hong: The Design of sites – Patterns for creating Winning web sites, Prentice Hall, 2nd Edition.
4. Adobe Creative Team: Action Script 3.0 - Adobe Flash Professional CS5 Classroom in book.

Web References:

1. [http://www.techdocs.ku.edu/docs/web-authoring_design-\[basics.pdf\]](http://www.techdocs.ku.edu/docs/web-authoring_design-[basics.pdf])
2. <https://www.youtube.com/watch?v=IjM9p0ISGZo> – [Photoshop tutorial]
3. <http://mashable.com/2010/08/12/12-beginner-tutorials-for-getting-started-with-photoshop/> - [Overview of tools]
4. <http://www.photoshopessentials.com/photo-effects/> - [Photo effects]
5. <http://www.1stoptutorials.com/Flash-Beginners-Tutorial-Course.html> - [Introduction to Flash]
6. <http://www.freeadobeflashtutorials.com/> - [Flash tutorial]

Course Outcomes: Upon successful completion of the course:

1. C01 : Students will learn basics of Photoshop Tool.
2. C02 : Students will familiar about advanced concepts of web designing such as Masking, Animation and Filtering.
3. C03 : Students will learn basics of Flash tool.
4. C04 : Students will able to design animations using Flash.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	3	1	1	2	1
CO2	3	3	2	2	1	1	2	1
CO3	3	2	3	2	1	1	2	1
CO4	3	2	3	2	1	1	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS317: Web Development Using Advanced Java (200 Marks)

Credit Hours: 07

Pre-requisite: None.

Methodology & Pedagogy: During theory sessions topics related to technologies and techniques used in the development of Web-based applications. During Practical sessions, students will be required to develop Applications using the concepts of JDBC, JSP, JSTL and Servlets. Student will also be explored to MVC architecture.

Learning Outcomes: Upon successful completion of the syllabus students shall be able to understand the concepts of web technologies and apply web technologies concepts to develop web applications.

Outline of the Course:

Unit No.	Title of the Unit	Minimum No. of Hours	
		Theory	Practical
1	Database Programming	10	36
2	Java Web Application Components	09	
3	Java Server Pages	08	
4	JSP Standard Tag Library	05	
5	Working with Servlets	10	
6	Advanced Servlet Features & Security	06	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit – I: Database Programming Hours: 10

The Design of JDBC, The Structured Query Language, JDBC Configuration, Executing SQL Statements, Query Execution, Scrollable and Updatable Result Sets, Row Sets, Metadata, Transactions

Unit – II: Java Web Application Components Hours: 09

Understanding Web Applications: Understanding Web Components, Servlets (Overview), JSP Pages, Introducing the MVC Design Pattern, Understanding Containers, Packaging Web Applications, Web Application Structure, JAR Files, WAR Files, HTTP, GET Request Method, POST Request Method, GET & POST in HTML Form Processing, Other Request Methods, The HTTP Response, Using Deployment Descriptors

Unit – III: Java Server Pages Hours: 08

Introducing JSP Technology – Understanding the Page Life Cycle, JSP Documents, JSP Document Syntax, A Simple JSP Document, HTTP, HttpJspPage Interface.

Introducing JSP Syntax and Usage – General Rules of Syntax, JSP Elements, Available Object Scope, Implicit Objects, Scripting Elements, Comments, JSP Expression Language

Unit – IV: JSP Standard Tag Library Hours: 05

Introduction to JSTL, Core Tag Library

Unit – V: Working with Servlets Hours: 10

Working with Servlets: Introducing Servlet, Introducing Servlet & the MVC Pattern, Introducing javax.servlet Package, Introducing the Servlet Interface, Introducing the GenericServlet Class, Introducing HTTP & Servlets, Understanding the Request/ Response Cycle, Input & Output Streams, Introducing Servlet/ Container Communication, Introducing ServletContext, Understanding the Deployment Descriptor, Introducing ServletContext Lifecycle Classes, RequestDispatcher Interface, Using Filters & RequestDispatcher

Unit – VI: Advanced Servlet Features & Security Hours: 06

Understanding the Stateless nature of HTTP, Why Track Client Identity & State? Maintain Sessions, Session Management Using the Servlet API, and Concepts of Filters.

Core Books:

1. Cay S Horstmann, Gary Cornell: Core Java, Volume II – Advanced Features, 8th Edition, Pearson Education.
2. Sue Spielman and Meeraj Kunnumpurath: Pro J2EE 1.4, Wiley Computer Publishing, 2004.
3. Marty Hall, Larry Brown: Core Servlets and JavaServer Pages, Volume 2, Advanced Technologies, 2nd Edition, Pearson Education, 2008.

Reference Books:

1. Marty Hall, Larry Brown: Core Servlets and JavaServer Pages, Volume 1, Core Technologies, 2nd Edition, Pearson Education, 2003.

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2. Bryan Basham, Kathy Sierra and Bert Bates: Head First Servlet and JSP, O'Reilly Publication, 1st Edition.

Web References:

1. <http://courses.coreservlets.com/Course-Materials/csajsp2.html> [Servlet Basics]
2. <http://www.ceit.es/asignaturas/IntelInfo/Recursos/Servlets/JavaServlets.pdf> [Servlet Tutorial PDF]
3. <http://www.msuniv.ac.in/AdvancedJavaProgrammingwithDatabaseApplication.pdf> [JDBC Tutorial]
4. www.doc.ic.ac.uk/~rcheung/teaching/2720/ppt/lecture12.ppt [JSP Tutorial Slides]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Be familiar with Enterprise Edition Framework of Java
C02 : Be familiar with web application development using Servlet
C03 : Be familiar with web application development using JSP
C04 : Able to create web application using advance Java
C05: Able to develop MVC application using Java enterprise edition

Course Outcomes Mapping:

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Database Programming				✓	✓
2	Java Web Application Components	✓	✓	✓	✓	✓
3	Java Server Pages	✓	✓	✓		
4	JSP Standard Tag Library	✓		✓	✓	✓
5	Working with Servlets	✓	✓		✓	✓
6	Advanced Servlet Features & Security	✓			✓	✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	1	2	2	1	2	1
CO2	3	3	2	2	2	1	2	1
CO3	3	3	2	2	2	1	2	1
CO4	3	3	3	-	-	-	3	-
CO5	3	3	3	-	-	-	3	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS318: Web XML Applications (200 Marks)

Credit Hours: 07

Pre-requisite: None.

Methodology & Pedagogy: During theory sessions concepts related to Service Oriented Architecture and concept of software as a service will be discussed using proper example. Emphasis will be given on concepts of service management using WCF service architecture, standards, frameworks and measurement aspects.

Learning Outcomes: Upon successful completion of the course, the students shall be able to create, host and use the WCF services.

Outline of the Course:

Unit No.	Title of the Unit	Minimum No. of Hours	
		Theory	Practical
1	Introduction to WCF	08	36
2	Understanding WCF	09	
3	Programming WCF	09	
4	WCF transactions and security	08	
5	Customization, Interpretability and integration of WCF	08	
6	Deploying WCF	06	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit – I: Introduction to Windows Communication Foundation Hours: 08

Evolution of WCF, Evolution of Service Oriented Architecture (SOA), What is WCF, Where does WCF Services fit in, WCF architecture, The makeup of WCF, WCF features, WCF Concepts, Messages, Channels, Services, Behaviors, WCF programming model, Installing WCF, Creating your first WCF service.

Unit – II: Understanding Windows Communication Foundation Hours: 09

Addresses, Understanding and programming WCF addresses, Understanding and programming WCF bindings, Understanding and programming WCF contracts

Unit – III: Programming Windows Communication Foundation Hours: 09

Clients, client communication patterns, creating client code, Services, Service's types, contracts and end points, Services behavior, Handling WCF Exceptions/Faults

Unit – IV: WCF transactions and security Hours: 08

Transactions overview, Transactions attribute in system and service model, Reliable sessions, Message exchange, security messages, Queues, Security overview, Security behaviors and bindings, Security clients and services

Unit – V: Customization, Interpretability and integration of WCF Hours: 08

Extending Service host and service model layer, Extending channel layer, Extending bindings, Interpretability and integration

Unit – VI: Deploying Windows Communication Foundation Hours: 06

Installing WCF services, Upgrading services, Managing WCF, Tracing, message logging, Service Configuration Editor, Hosting WCF services

Core Books:

1. Scott Klein: Professional WCF Programming, WROX Publication, April 2007.
2. Craig McMurty, Marc Mercuri, Nigel Watling, Matt Winkler: Windows Communication
3. Foundation Unleashed, SAMS Publication, March 2007.
4. Microsoft Press – Step By Step of WCF.

Reference Books:

1. Michele Leroux Bustamante: Learning WCF - A Hands-on Guide, 1st Edition, O'Reilly Media.
2. Steve Resnick, Richard Crane, Chris Bowen: Essential Windows Communication Foundation for .NET Framework 3.5, 1st Edition, Addison-Wesley Professional.

Web References:

1. www.wcftutorial.net/ [Lecture Notes and Practical Demo]
2. www.tutorialspoint.com/wcf/ [Lecture Notes and Practical Demo]
3. www.topwcftutorials.net/ [Lecture Notes and Practical Demo]
4. <http://www.codeproject.com/Articles/406096/A-beginners-tutorial-for-understanding-%20Windows> [Introduction of WCF]
5. <http://java.sun.com/webservices/interop/index.jsp> [To understand Interpretability]

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Course Outcomes: upon successful completion of the course, the student will:

CO1: Understand the basics of service oriented software development architecture.

CO2: Identify different service oriented application development technologies

CO3: Identify strengths and limitations of different service oriented software development technologies.

CO4: Develop an understanding of WCF technology.

CO5: Be able to solve real world challenges by developing applications based in WCF technology and apply it in suitable domains.

CO6: Be able to debug WCF applications with inbuilt tools and deploy in different ways.

Course outcome mapping:

Unit No.	Unit Name	Course Outcome					
		C01	C02	C03	C04	C05	C06
1	Introduction to WCF	✓	✓				
2	Understanding WCF			✓			
3	Programming WCF				✓		
4	WCF Transactions & Security				✓		
5	Customization, Interpretability and Integration of WCF					✓	
6	Deploying WCF						✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	2	-	-	2	-	1	-	-
CO2	3	2	-	-	-	-	3	-
CO3	-	3	-	2	-	-	-	-
CO4	-	1	-	-	-	-	1	-
CO5	-	-	-	-	-	1	-	-
CO6	-	-	2	2	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

MS319: Web Development Using Open Source (200 Marks)

Credit Hours: 07

Pre-requisite: None.

Methodology & Pedagogy: In order to achieve the course objectives, students will be introduced to the open source concepts, various products like linux / windows, apache, mysql and php will be covered. The session will cover the Installation and use of PHP on a local server, Utilizing decision making constructs in a PHP program, Utilizing techniques to overcome the statelessness of the Web, Using file I/O to read and write to files, Installing mySQL, Using phpMyAdmin, and exploring PEAR.

Learning Outcomes: Upon successful completion of the syllabus students shall be able to understand, use and develop applications using open source software.

Outline of the Course:

Unit No.	Title of the Unit	Minimum No. of Hours	
		Theory	Practical
1	Introduction to open source software and their configuration	06	36
2	Introduction to PHP	08	
3	HTML forms with PHP	09	
4	Accessing Mysql with PHP	10	
5	Introduction to Object Oriented Programming and Web Features	08	
6	PHP Utilities and Understanding PEAR	07	

Total Hours (Theory): 48

Total Hours (Lab): 36

Total: 84

Detailed Syllabus:

Unit – I: Introduction to open source software and their Configuration Hours: 06

Overview of Open Source Software, Widely used open source products, Development philosophy, Open source vs. closed source, Open source software vs. free software, Open

source vs. source-available, Pros and cons, Development tools, The distribution terms of open source software, open source technology importance, Free and Open Source Software (FOSS), Configuring apache, Configuring Mysql, Configuring PHP.

Unit – II: Introduction to PHP Hours: 08

Overview of PHP structure and syntax: Evolution of PHP, basic rules of php syntax, using variables, data types, operators and expressions, decisions and loops, Strings, Arrays and Functions.

Unit – III: HTML forms with PHP Hours: 09

Working of HTML with PHP, Capturing data with PHP Using radio buttons, checkbox, list box, buttons, text box, etc., processing user input, Handling and Avoiding errors, Exception Handling, working with regular expressions (regex), Understanding files and directories, operations on files and directories.

Unit – IV: Accessing Mysql with PHP Hours: 10

Overview of Mysql structure, Connectivity, Querying the database, Retrieving and Sorting, Restricting Data, MySQL Functions, Displaying Data from Multiple Table (Joining), Aggregate Functions and Aggregation Using GROUP BY Clause, Sub query, manipulating table, Creating and Managing Table.

Unit – V: Introduction to Object Oriented Programming and Web Features Hours: 08

Instantiation, Modifiers, Inheritance, Interfaces, Exceptions, Static Methods and Properties, Auto load, Reflection, Type Hinting and Class Constance. Preserving state of objects with query string, working with cookies, using Php session to store data.

Unit – VI: PHP Utilities and Understanding PEAR Hours: 07

Date-Time functions, checking and formatting date values, Timestamps, Understanding HTTP requests, Exploring and modifying HTTP responses, Getting information from web server, Sending mails, Introduction to PEAR, Installing and using PEAR package.

Core Books:

1. Matt Doyle: Beginning PHP 5.3, Wrox, 2010.
2. Hugh E. Williams, David Lane: Web Database Applications with PHP and MySQL, Building Effective Database-Driven Web Sites, 2nd Edition, O'Reilly Media, May 2004.

Reference Books:

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1. Timothy Boronczyk, Elizabeth Naramore, Jason Gerner, Yann Le Scouarnec, Jeremy Stolz: Beginning PHP6, Apache, and MySQL Web Development, Wrox, 2010.
2. Jason Gerner, Morgan Owens, Elizabeth Naramore, Matt Warden: Professional LAMP Linux, Apache, MySQL and PHP Web Development, Wiley India, 2006.

Web References:

1. <http://www.cecs.csulb.edu/~mopkins/cecs470/PHPNotes.pdf> [PHP Introduction]
2. www.university.youth4work.com › Study Material [PHP Tutorials]
3. <http://freevideolectures.com/Course/2331/Building-Dynamic-Websites/2> [PHP Tutorials]

Course Outcomes: Upon successful completion of the course, the students will:

- C01 : Understand the concepts of Open Source Software.
C02 : Be able to work with PHP and design webpages using HTML.
C03 : Understand database integration and creation in the web application development.
C04 : Be familiar with OOP in PHP
C05: Be able to understand PEAR.

Unit No.	Unit Name	Course Outcomes				
		C01	C02	C03	C04	C05
1	Introduction to open source software and their configuration	✓				✓
2	Introduction to PHP		✓	✓		
3	HTML forms with PHP		✓	✓		
4	Accessing MySQL with PHP		✓	✓	✓	
5	Introduction to Object Oriented Programming and Web Features			✓	✓	
6	PHP Utilities and Understanding PEAR					✓

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
C01	3	-	-	2	-	-	3	-	2
C02	2	2	-	-	-	-	3	-	-
C03	-	3	2	-	-	2	-	-	-
C04	3	-	2	-	2	-	2	2	2
C05	2	3	-	2	3	3	2	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

OBJECTIVES, TEACHING SCHEME & DETAILED SYLLABUS

FOR

**M.Sc. (IT) PROGRAMME
(4th SEMESTER)**

**EFFECTIVE FROM
ACADEMIC YEAR 2018-19**

Faculty of Computer Science and Applications

SEMESTER-IV

Course Code	Course Title	Teaching Scheme			Credit	Internal Continuous Evaluation	End Semester Examination		Total
		Contact Hours					Report	Presentation & Viva	
		Inst.	Industry	Total					
MS401	Dissertation / Project Work	2	28	30	30	200	200	400	800

MS401: Dissertation / Project Work

(800 Marks)

Guidelines to Undertake the Final Semester M.Sc.(IT) Dissertation/ Project

Project/Dissertation Work:

Students of M.Sc. IT have to do the Project/Dissertation work in an Industrial/ Research Organization of computer field. Project/Dissertation work shall be carried out under the supervision of a qualified teacher in the Department as well as an expert from organization. Students have to meet the institute supervisor periodically and to attend the project/dissertation review meetings for evaluating the progress. The Dissertation/Project work shall be pursued for a minimum of 16 weeks during the semester.

MORI Principle to choose the topic:

Final semester dissertation/project work is important and challenging element of M.Sc. IT study. Selection of topic of project/dissertation work is very crucial and important aspect of that. Following are important principles that will help students to select the appropriate topic of project/dissertation work.

- **Manageable:** your dissertation topic must be sufficiently focused so that it is possible for you to do the topic justice within the available time (one semester). You may have a real interest in, say, *'the impact of technology computers on Indian Economic Growth since Independence'*, but you certainly won't be able to cover this topic in any detail in the space of one semester (or around 5-7 thousand words)!
- **Original:** this relates to the above point, since a topic that is focused and manageable is more likely to be one that has not been written about too extensively, thus leaving room for your original contribution. Ideally you will find an interesting and well-chosen topic which will impress those marking your work.
- **Relevant:** your project should clearly be relevant to some aspect of your studies, but it might also be relevant to your plans for, say, postgraduate study or a career. The dissertation may also be relevant in the sense that it plays to some of your established strengths, such as a particular course module or topic that you have enjoyed studying and in which you have previously done well.
- **Interesting:** you are obviously more likely to enjoy and be successful in your dissertation if it is of real interest to you and to those marking your work. Ask yourself if you are sufficiently committed to your idea to be able to give it your best throughout the duration of your project. You should also ascertain whether your supervisor finds the idea interesting during your initial discussions with her or him.

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Evaluation Of Project/Dissertation Work:

The dissertation / project report shall normally be written in English in the specified format and shall be characterized by significant contribution to knowledge in the field. Normally two copies of the report are to be submitted for evaluation. The Project/Dissertation report prepared according to approved guidelines and duly signed by the supervisor(s) and the Head of the Department shall be submitted to the Head of the Institution. The evaluation scheme of Project/Dissertation is as under:

Course	Course Title	Teaching Scheme			Credit	Internal	End Semester Examination		Total
		Contact Hours		Continuous Evaluation		Report	Presentation & Viva		
		Inst.	Industry					Total	
MS401	Dissertation/Project Work	2	28	30	30	200	200	400	800

The internal evaluation of project/dissertation is done based on progress reports and internal presentations.

The final evaluation of the project/dissertation will be based on the project report submitted and a Viva-Voce Examination by a Board of Examiners.

If a candidate fails to submit the project report on or before the specified deadline, he/she is deemed to have failed in the Project /Dissertation Work and shall re-enroll for the same in a subsequent semester. If a candidate fails in the viva-voce examinations of Project/ Dissertation work, he/she shall resubmit the project report within specified duration decided by university. The resubmitted project will be evaluated during the subsequent academic session. A copy of the approved project report after the successful completion of viva examinations shall be kept in the library of the college / institution.

Web References:

1. <http://www.microtoolsinc.com/Howsrs.php> [For Software Requirement Specification Guidelines]
2. <http://techwhirl.com/writing-software-requirements-specifications/> [For goals of SRS]
3. <https://kepler-project.org/developers/reference/software-development-guidelines> [For Software Development Guidelines]

Course Outcomes :

CO1 : Student will understand the implementation of concepts of SDLC and Software Engineering.

CO2 :The programming concepts they learn during their academics, it will be converted in to the actual implementations.

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CO3: Students will be exposed to understand the requirement of proposed software and implement these requirements in terms of programming logic and methods.

CO4: Students must understand the difference between a program and professional application/product/software.

CO5: Students will learn different categories of applications like Desktop application, Web applications, etc.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	3	3	1	2	2	3	2
CO2	3	3	3	1	2	2	3	2
CO3	3	3	3	1	2	2	3	2
CO4	3	3	3	1	2	2	3	2
CO5	3	3	3	1	2	2	3	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”